

Chapter 6
Assignment 5

Deadline: March 10, 2026

Short Answer Questions:

1. Define Full-Stack Development.
2. Differentiate between Frontend and Backend development.
3. What is Web Application Deployment?
4. Define Testing and Quality Assurance (QA).
5. What is the difference between Unit Testing and Integration Testing?
6. What is the difference between Continuous Delivery and Continuous Deployment?
7. What is DevOps?
8. Mention any two benefits of DevOps practices.
9. What is meant by Usability in web applications?
10. What is the purpose of a Service Worker in PWA?

Long Answer Questions:

1. Explain Full-Stack Development architecture with a neat diagram. Describe the interaction between client, server, and database.
2. Discuss different types of software testing used in web applications. Explain:
 - Unit Testing
 - Integration Testing
 - System Testing
 - User Acceptance Testing
3. Explain the concept of DevOps. How does it improve collaboration between development and operations teams?
4. Describe the complete CI/CD pipeline process from code commit to deployment. Illustrate with a real-world example.
5. What are Progressive Web Apps (PWAs)? Explain their key features such as:
 - Offline support
 - Installability
 - Push notifications
 - Service workers

6. Explain Responsive Web Design. Discuss techniques such as:
 - Media queries
 - Flexible grids
 - Flexible images
 - Viewport meta tag
7. Discuss the importance of usability in modern web applications. Explain usability principles such as:
 - Accessibility
 - Simplicity
 - Consistency
 - User feedback
8. Write short notes on following modern trends in web deployment:
 - Cloud hosting
 - Containerization
 - Microservices
 - Serverless architecture
 - Cloud Databases